

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 31-08-2026

TITLE

Design Static Webpage Using HTML Components

AIM

To understand and implement pure semantic HTML5 components for designing a static webpage without using any CSS styling.

OBJECTIVES

- Create a valid HTML5 document using only semantic and structural tags.
- Design a page header with a title and navigation links using tables and native elements.
- Organize page content using semantic elements such as main, article, section, and aside.
- Build an interactive demonstration form using native HTML form controls.
- Apply ordered and unordered lists to present structured information.

TASK DESCRIPTION

Design a pure semantic HTML webpage titled "StaticWeb" that relies entirely on native browser defaults for layout and presentation, with no external or internal CSS. The page includes a header with a site title and navigation bar, a hero/showcase section, a two-column main layout built with a structural table (an article column and a sidebar), a demonstration inquiry form using standard form controls, and a footer.

ALGORITHM/PROCEDURE

1. Create an HTML5 document using the DOCTYPE declaration and html, head, and body elements.
2. Add a header section containing the site title and a navigation bar using a table for alignment.
3. Add a hero/showcase section introducing the purpose of the page.
4. Create a two-column main layout using a structural table: a left column for main content and a right column for a sidebar.
5. Inside the main column, add article elements describing core structural tags and native interaction elements.
6. Design a demonstration inquiry form using fieldset, legend, text, email, select, and textarea controls.
7. Inside the sidebar (aside), add unordered and ordered lists describing benefits and required components.
8. Add a footer with closing text.
9. Open the HTML file in a browser and verify the displayed webpage.

PROGRAM/SOURCE CODE

The following represents the completely formatted HTML source code, organized with clean indentation and descriptive comments:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>

  <!-- HEADER SECTION: Contains the site title and primary navigation -->
  <header>
    <!-- Using a table for structural layout to align title left and
    nav right -->
    <table width="100%" border="0" cellspacing="0" cellpadding="10">
      <tr>
        <td>
          <h1>StaticWeb</h1>
        </td>
        <td align="right">
          <!-- Navigation bar with internal anchor links -->
          <nav>
            <a href="#home">Home</a> &nbsp; &nbsp; &nbsp;
            <a href="#articles">Articles</a> &nbsp; &nbsp; |&nbsp; &nbsp; &nbsp;
```

```

        <a href="#about">About</a> &nbsp;   </a>
        <a href="#contact">Contact Us</a>
    </nav>
</td>
</tr>
</table>
</header>
<hr>

<!-- HERO / SHOWCASE SECTION: Introductory banner area using native
centering -->
<section id="home">
    <center>
        <h2>Semantic Structures Without CSS</h2>
        <p>This layout uses raw HTML components to build an organized
document hierarchy that browsers can cleanly parse out-of-the-box.</p>
        <p>
            <a href="#articles"><b>Read Our Articles Below
&darr;</b></a>
        </p>
    </center>
</section>
<hr>

<!-- MAIN CONTENT LAYOUT: A structural table to create a 2-column
layout -->
<table width="100%" border="0" cellspacing="0" cellpadding="20">
    <tr valign="top">

        <!-- LEFT SIDE: Main Content Column (70% width) -->
        <td width="70%">
            <!-- The <main> tag denotes the dominant content of the
document -->
            <main id="articles">

                <!-- First Article block -->
                <article>
                    <h2>1. Core Structural Tags</h2>
                    <p>Published: July 7, 2026</p>
                    <p>When you omit cascading style sheets entirely,
the semantic markup handles all of the structural heavy lifting. Elements
like <code>&lt;main&gt;</code>, <code>&lt;article&gt;</code>, and
<code>&lt;section&gt;</code> tell screen readers and web scrapers exactly
how information flows.</p>
                    <p>Using raw browser defaults ensures excellent
loading performance, perfect accessibility compliance, and absolute
structural clarity.</p>
                </article>

                <br><br><br>

                <!-- Second Article block with interactive form -->

```

```

<article>
    <h2>2. Native Interaction Elements</h2>
    <p>Published: July 5, 2026</p>
    <p>Browsers come equipped with excellent
interactive elements by default. We can capture text inputs, provide
multiple-choice configurations, and submit structured information to
backend parsers using pure markup.</p>

    <!-- FORM SECTION: User interaction demonstration -
->

    <h3>Interactive Demonstration Forms</h3>
    <form id="contact" action="#" method="post">

        <!-- Fieldset groups related form elements and
draws a box around them -->
        <fieldset>
            <legend><b>User Inquiry Form</b></legend>

            <p>
                <label for="name">Full Name: </label>
                <!-- Text input for capturing user's
name -->
                <input type="text" id="name"
name="username" required placeholder="John Doe">
            </p>

            <p>
                <label for="email">Email Address:
</label>
                <!-- Email input with built-in browser
validation -->
                <input type="email" id="email"
name="useremail" required>
            </p>

            <p>
                <label for="topic">Inquiry Topic:
</label>
                <!-- Dropdown selection menu -->
                <select id="topic" name="usertopic">
                    <option value="general">General
Feedback</option>
                    <option value="technical">Technical
Architecture</option>
                    <option value="academics">Academic
Curriculum</option>
                </select>
            </p>

            <p>
                <label for="message">Message Details:
</label><br>

```

```

                                <!-- Multi-line text area for long
inputs -->
                                <textarea id="message"
name="usermessage" rows="5" cols="40" placeholder="Type your notes
here..."></textarea>
                                </p>
                                <p>
                                <!-- Form submission and reset buttons
-->
                                <input type="submit" value="Submit Form
Data">
                                <input type="reset" value="Clear
Entries">
                                </p>
                                </fieldset>
                                </form>
                                </article>
                                </main>
                                </td>

                                <!-- RIGHT SIDE: Sidebar Column (30% width) -->
                                <td width="30%" bgcolor="#f4f4f4">
                                <!-- The <aside> tag is used for content tangentially
related to the main content -->
                                <aside id="about">
                                    <h3>Document References</h3>
                                    <p>Static pages map data out directly without real-time
database queries or processor overhead.</p>

                                    <!-- Unordered (bulleted) list -->
                                    <h4>Core Benefits:</h4>
                                    <ul>
                                        <li>Instant browser painting</li>
                                        <li>Low memory footprints</li>
                                        <li>High data compatibility</li>
                                    </ul>

                                    <!-- Ordered (numbered) list -->
                                    <h4>Required Components:</h4>
                                    <ol>
                                        <li>Document Type Definition</li>
                                        <li>Semantic Wrapper Blocks</li>
                                        <li>Data Input Control Interfaces</li>
                                    </ol>
                                </aside>
                                </td>
                                </tr>
                                </table>
                                <hr>

```

```

    <!-- FOOTER SECTION: Copyright and closing information -->
    <footer>
        <center>
            <p>&copy; 2026 Pure HTML Architecture. Assembled completely
without style configurations.</p>
        </center>
    </footer>

</body>
</html>

```

CODE EXPLANATION

HTML Element / Feature	Purpose & Explanation
HTML5 DOCTYPE	Declares that the document uses HTML5.
header / h1	Displays the site title StaticWeb.
table (header)	Aligns the site title and navigation links on the same row.
nav / a	Provides navigation links to Home, Articles, About, and Contact Us.
section#home / center	Presents the hero showcase heading and introduction text.
table (main layout)	Creates a two-column structural layout without CSS.
main / article	Organizes core content into logically grouped articles.
code	Displays inline HTML tag names as formatted code text.
form / fieldset / legend	Groups the inquiry form and labels it.
input (text, email)	Collects the user's name and email address.
select / option	Provides a dropdown to choose the inquiry topic.
textarea	Accepts longer free-text message details.
input (submit, reset)	Submits or clears the form data.
aside	Displays supplementary sidebar content.
ul / ol / li	Presents core benefits and required components as lists.
footer	Displays the closing copyright text.

TEST CASES AND EXECUTION DATA

Test Case	Verification	Expected Result	Observed Result	Status
TC-01	Page loading	Webpage is	Page displayed	Pass

		displayed		
TC-02	Navigation	Navigation links are visible	Links visible	Pass
TC-03	Layout structure	Two-column layout is displayed	Layout displayed correctly	Pass
TC-04	Inquiry form	Form controls are displayed	Form controls displayed	Pass
TC-05	Sidebar lists	Benefit and component lists are visible	Lists visible	Pass

OUTPUT

The following browser output screenshots depict the execution of the 5 Test Cases based on the implemented HTML code:

TC-01: Page loading (Webpage is displayed cleanly)

StaticWeb

[Home](#) [Articles](#) | [About](#) [Contact Us](#)

Semantic Structures Without CSS

This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.

[Read Our Articles Below ↓](#)

TC-02: Navigation links are visible

StaticWeb

[Home](#) [Articles](#) | [About](#) [Contact Us](#)

TC-03: Layout structure (Two-column layout is displayed using a table)

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like <main>, <article>, and <section> tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

Document References

Static pages map data out directly without real-time database queries or processor overhead.

TC-04: Inquiry form (Form controls are displayed natively)

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

General Feedback

Message Details:

Type your notes here...

Submit Form Data

Clear Entries

TC-05: Sidebar lists (Benefit and component lists are visible)

Core Benefits:

- Instant browser painting
- Low memory footprints
- High data compatibility

Required Components:

1. Document Type Definition
2. Semantic Wrapper Blocks
3. Data Input Control Interfaces

RESULT

Thus, a static webpage was successfully designed using pure semantic HTML components without any CSS styling. The webpage demonstrates HTML5 document structure, semantic layout elements, table-based alignment, navigation links, an interactive demonstration form, and structured lists. The supplied browser screenshots verify the implemented features.